SANDIA REPORT
SAND20XX-XXXX
Printed November 2021

Sandia National Laboratories

MANIPULATE-2020: An Interface Tool for Nuclear Data Processing

Patrick J. Griffin

Prepared by
Sandia National Laboratories
Albuquerque, New Mexico
87185 and Livermore,
California 94550

ABSTRACT

LIST OF FIGURES

LIST OF TABLES

ACRONYMS AND DEFINITIONS

Abbreviation	Definition
DOE	Department of energy
ENDF	Evaluated Nuclear Data File
ESTSC	Energy Science and Technology Software Center
NEA	Nuclear Energy Agency
NTESS	Nuclear Technology and Engineering Solutions of Sandia
SNL	Sandia National Laboratories
WSL	Windows Subsystem for Linux

1. PURPOSE

2. CODE RELEASE DOCUMENTATION

2.1. Software Name

2.2. Software Version

2.3. Author(s)

2.4. Software Completion Date

2.5. Brief Description

2.6. Practical Application

2.7. Method of Solution

2.8. Computer(s) and Operating System for Which Software is Written

2.8.1. Programing Language

2.9. Compiler Version

2.10. Software Limitations

2.11. Related/Auxiliary/Co-Requisite Software

2.12. Hardware Requirements

2.13. References

2.14. Funding Information

2.15. Treaties/Agreements/IP Management Plans

2.16. Sensitivities

2.17. Commercialization/Distribution Plans

https://github.com/pjgriff/Manipulate-2020

https://gitlab-ex.sandia.gov/bdhehr/Manipulate-2020

2.18. Further Development

3. INSTALLATION

4. DESCRIPTION OF STRUCTURE OF INPUT FILE

Table 4-1. Control Flags for Manipulate-2020 Code

Control Flags: format 19I2
c icon(1) type of action
c = 9 end
c = 1 combine cross section elements - logic unclear, PJG 10/17/2012
c = 2 prepare component plot file format output
c = 3 display a matxsr file from njoy
c =-3 same as 3 for xxx.plt 89 file
c = 4 multiple plot file option
c = 5 sum and plot components
c =-5 perform statistical ananlysis on a series of functions
c = 6 fold source and response
c = 7 weight and plot function
c =-7 reciprocal weight and plot function
c = 8 file difference
c =-8 covariance manipulation options
c =10 interpolate matxsr file
c =11 interpolate tabulated file
c =12 expand by interpolation a tabulated file
c =13 expand an analytic function in 640 group structure
c =14 combine covariance files in LSL format (common energy structure) - NEW
c =15 extract PKA recoil spectrum for given reaction/energy - NEW
c =16 extract integrated PKA recoil spectrum for as a function of energy - NEW
c
c icon(2) input energy grid/data format
c = 0 njoy sand-ii 620/640/725/770 point neutron cross section
c (energies in ev, response data to be multiplied
c by 10**-3 to yield kerma units of mev-mb)
c = 1 j. kelly sand-ii pun spectrum format
c (default quantity differential number spectrum)
c compute iselect material = 1 number fraction
c 2 energy fraction

Control Flags: format 19I2

```
c                                                    3 differential number fraction
c                                                    4 differential energy fraction
c                                                    5 integral number
c              = 2       same as 1
c              = 3       flash x-ray curr spectral input
c              = 4       list directed (plot output)
c              = 5       njoy sand-ii /770 point neutron pka recoil spectrum
c                        (energies in ev, response data to be multiplied
c                         by 10**-3 to kev)
c              = 6       njoy matxsr interface file - arbitrary structure
c              = 9       user defined - xxx.flux file
c              =10       640 group response structure = xxx.res
c              =11       arbitrary response structure = xxx.res
c
c         icon(3)    display/output special option
c              = 0       no action
c              = 1       punch in damout format
c              = 2       punch in tdam format
c
c         icon(4)    print energy order
c              = 0       punch high to low
c              = 1        punch low to high
c
c         icon(5)    sand structure
c              = 0        640 groups - SAND-II
c              = 1        770 groups - SAND-II extended
c              = 2         89 groups - NuGET neutron
c              = 3         48 groups - NuGET gamma
c              = 5        175 group - Vitamin-J
c              = 6        725 group - IAEA
c
c         icon(6)    inhibit renormalization
c              = 0       normal
c              = 1       response, inhibit renormalization in filein
c
c         icon(7)    NJOY interface location
c              = 0       sync Projects-Linux/NJOY-2016 default
c              = 1       sync Linux/NJOY-2016 selection
c
c         icon(9)    print inhibit
c             = 0        default print
c             =>0        inhibit print by this level
```

Control Flags: format 19I2

```
c             < 0        enhance print (debug option)
c
c         icon(10)   label replacement
c             = 0        no action
c             = 1        read new legend information
c
c         icon(11)   special action
c             = 0        not used
c             = 1        on pka plot, divide entry 1 by entry 3
c                        to produce average recoil energy -
c                        when icon(2) = 5
c             = 2        on icon(1)=7 combine, punch plt-format
c                        interface file on indicated dir:filename
c                        (tbd)
c          icon(12)   reverse order of data read - only partially implemented
c             = 0        no effect
c             = 1        reverse
c          icon(13)   attenuation function
c             = 1        if equal to 1 and icon(1)=6, then
c                        use new function for exponential attenuation
c                        with coefficient alpha
c             = 2        if equal to 2 and icon(1)=6, then
c                        use sensor self-shielding correction factor
c          icon(14)   new title
c             = 0        use first card in input stream
c             = 1        read-in new title
c
c          icon(15)   flip order of data for punch
c             = 0         do nothing
c             + 1        invert order
c
c         icon(16)    re-direct to look for pka files
c             = 0         do nothing
c             = 1         look in njoy/output-pka directory
c             = 2         look in NJOY9-2012 for groupr/pka files
c
c        icon(17)     point interpolation for grprin option
c             = 0         no, not used
c             = 1         yes, interpolate
c
c        icon(18)     precision control for sum/difference
c             = 0         no action
```

Control Flags: format 19I2
c = 1 force 0.0 response for differences < 1.e-4 of maximum element
c
c icon(19) inversion of output energy order
c = 0 no change
c = 1 invert order
c = 6 njoy matxsr interface file - arbitrary structure

5. CODE EXECUTION INSTRUCTIONS

6. SAMPLE APPLICATION CASES

6.1. NJOY/GROUPR Response Extraction

6.1.1. Description

6.1.2. Output

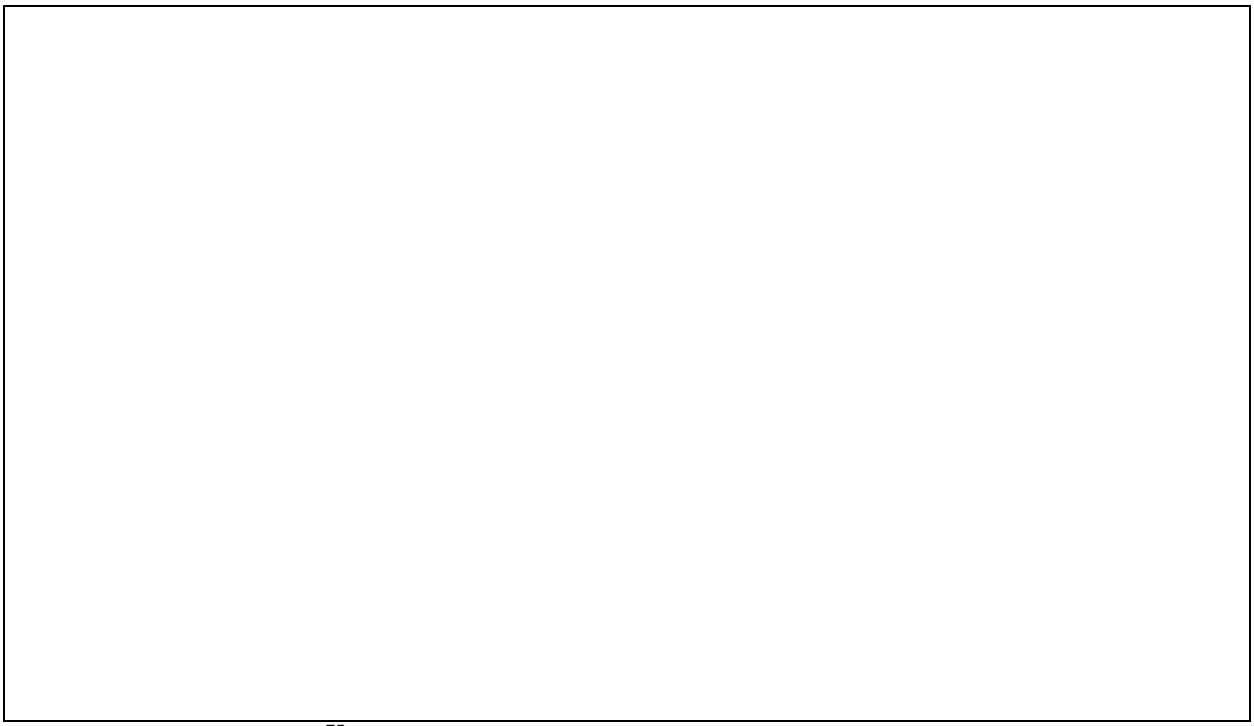


Figure 1: ^{75}As Neutron Total Kerma and Displacement Kerma Limit

6.2. Weight and Combine Response Functions

6.2.1. Description

6.2.2. Output

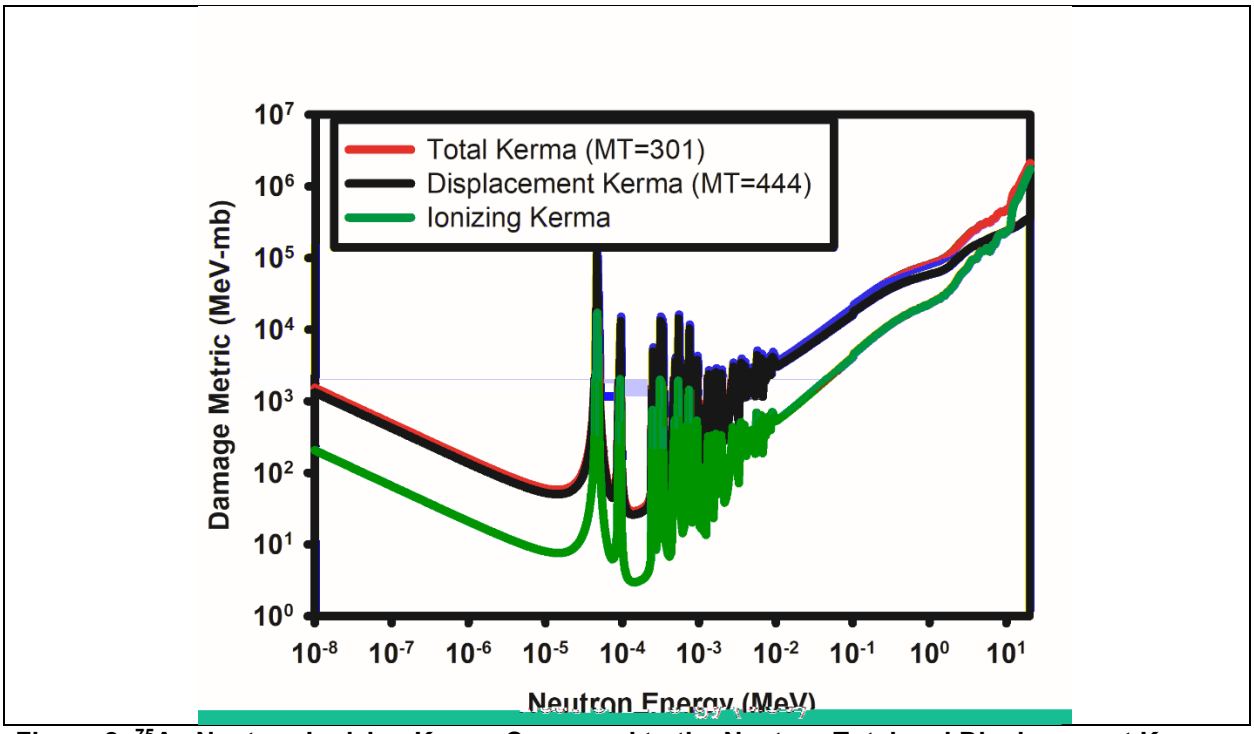


Figure 2: ^{75}As Neutron Ionizing Kerma Compared to the Neutron Total and Displacement Kermas

6.3. NJOY/GROUPR Cross Section Extraction

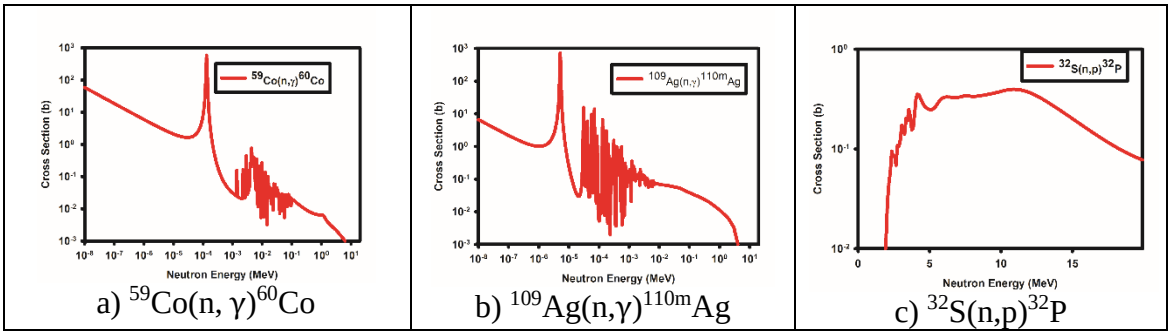
6.3.1. Description

γ

γ

γ

6.3.2. Output



a) $^{59}Co(n, \gamma)^{60}Co$ b) $^{109}Ag(n,\gamma)^{110m}Ag$ c) $^{32}S(n,p)^{32}P$

Figure 3: Cross Section for Example NJOY/GROUPR Extraction

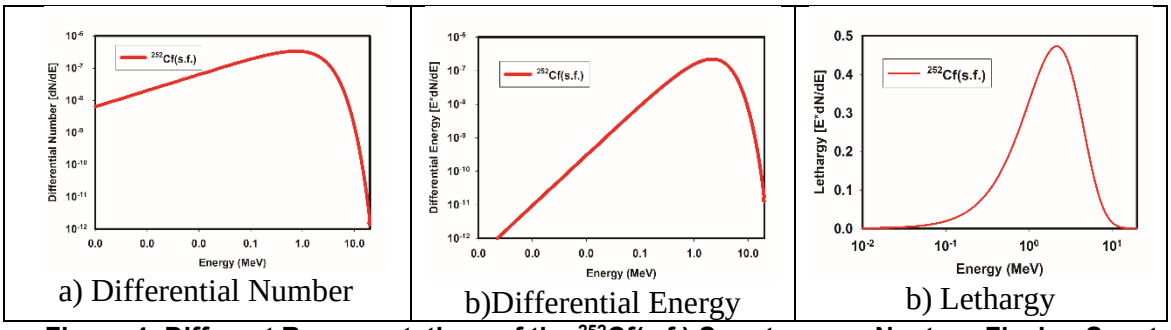
6.4. NJOY/GROUPR Spectrum Extraction

6.4.1. Description

6.4.2. Output

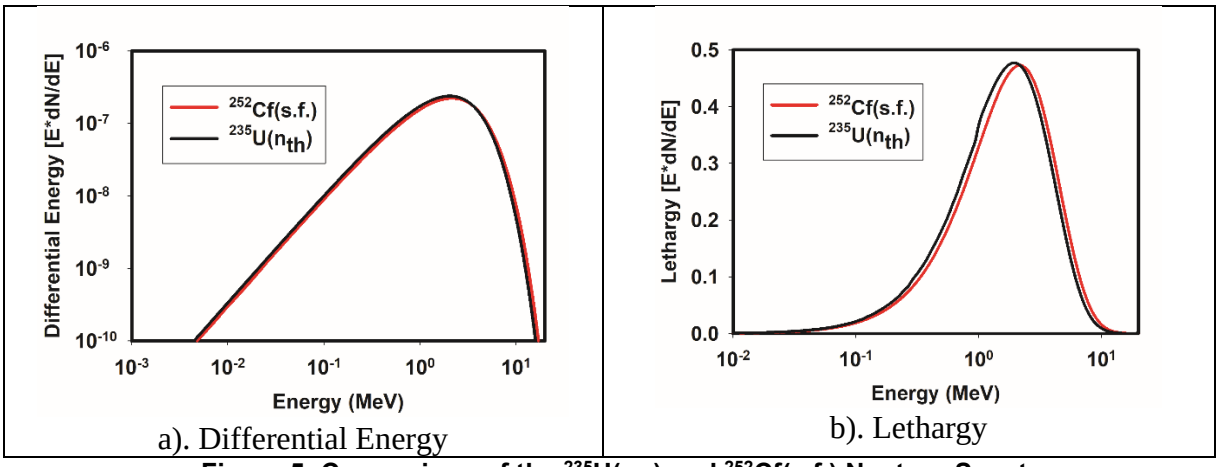
$\left\langle E_n^{252Cf(s.f.)} \right\rangle =$

$\left\langle E_n^{235U(n_{th})} \right\rangle =$



a) Differential Number b)Differential Energy b) Lethargy

Figure 4: Different Representations of the ^{252}Cf(s.f.) Spontaneous Neutron Fission Spectrum



a). Differential Energy b). Lethargy

Figure 5: Comparison of the ^{235}U(n_{th}) and ^{252}Cf(s.f.) Neutron Spectra

6.5. Folding of Cross Section with Neutron Spectrum

6.5.1. Description

γ γ

6.5.2. Output

Table 6-1. Summary Characteristic of Spectrum/Cross Section Folding

Cross Section	Neutron Spectrum	<σ> (mb)	Response Sensitivity Energy Limits (MeV)		
			5%	50%	95%
$^{59}Co(n,\gamma)^{60}Co$	$^{252}Cf(s.f.)$	4.864277	0.04768145	0.8829955	3.446157
	$^{235}U(n_{th})$	5.026129	0.04544459	0.8577079	3.286839
$^{109}Ag(n,\gamma)^{110m}Ag$	$^{252}Cf(s.f.)$	9.34751	0.07278118	0.7176283	2.615176
	$^{235}U(n_{th})$	9.739893	0.07204609	0.7047761	2.556543
$^{32}S(n,p)^{32}P$	$^{252}Cf(s.f.)$	73.99146	2.260848	4.024577	7.734848
	$^{235}U(n_{th})$	66.82598	2.235097	3.917196	7.298408

6.6. Validate Positive Definite Attribute for the Cross Section Covariance Matrix

6.6.1. Description

6.6.2. Output

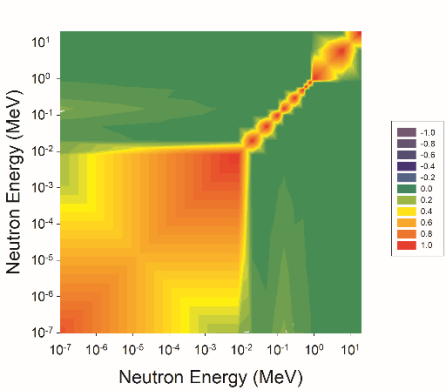
Table 6-2. Eigenvalues for Representative Dosimetry Cross Sections

Cross Section	Eigenvalues
$^{59}Co(n,\gamma)^{60}Co$	3.188861 0.2093912 0.6967719 0.8649985 0.9670087 0.500000 0.7999997 0.9729697 0.9000000 0.8999999
$^{109}Ag(n,\gamma)^{110m}Ag$	1.650150 0.3700236 1.097767 0.9657941 0.9778988 0.9821049 0.9854667 0.9950126 0.9937788 0.9899527 0.9920523 1.147018 0.8893262 0.9636558
$^{32}S(n,p)^{32}P$	13.95432 9.689728 6.911028 4.000117 1.946010 1.302666 1.249709 1.169561 1.231581 1.096308 0.9223984 0.7239132 0.5475745 0.6180022 0.4965166 0.1028905 0.4340180 0.2553883 0.3690853 0.3335642 0.2960616 0.3153051 0.7237343E-2 0.3198490E-2 0.3050713E-2 0.2880561E-2 0.1487948E-2 0.2106898E-2 0.2367189E-2 0.2635556E-2 0.2472797E-2 0.2558193E-2 0.9133368E-3 0.5724487E-3 0.4116856E-3 0.3343157E-6 0.3516820E-3 0.5826211E-4 0.2939752E-3 0.2508267E-3 0.2651053E-3 0.1056020E-3 0.2343758E-3 0.1228880E-3 0.1795922E-3 0.2119063E-3 0.1378094E-3 0.1313290E-3 1.000000

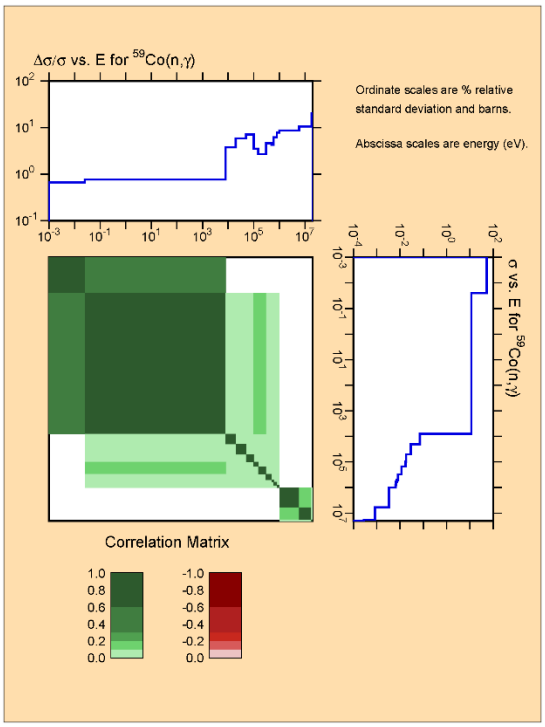
6.7. Cross Section Correlation Matrix Conversion for Plotting Interface

6.7.1. Description

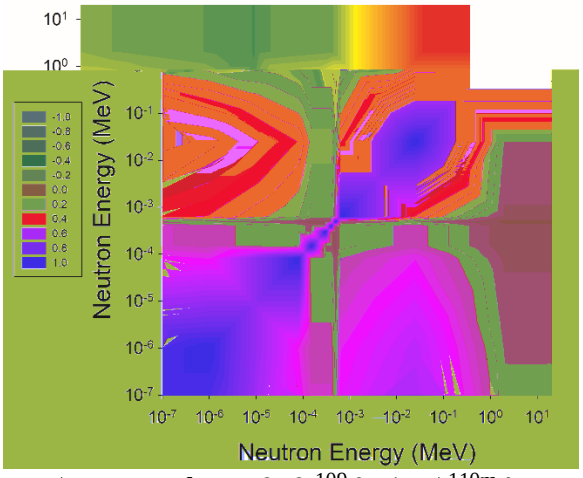
6.7.2. Output



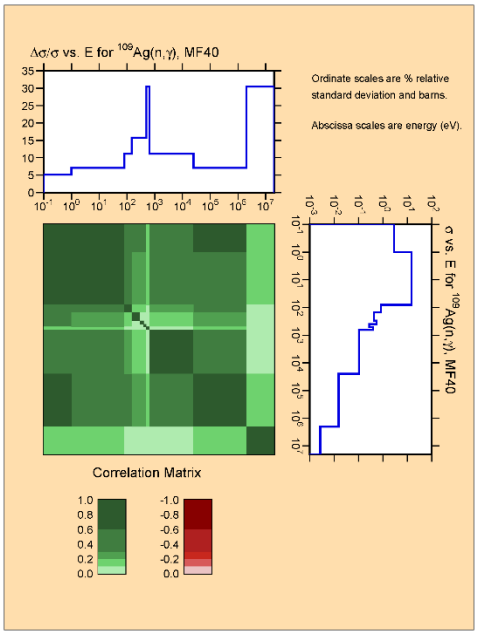
a). Manipulate-2020 $^{59}Co(n, \gamma)^{60}Co$



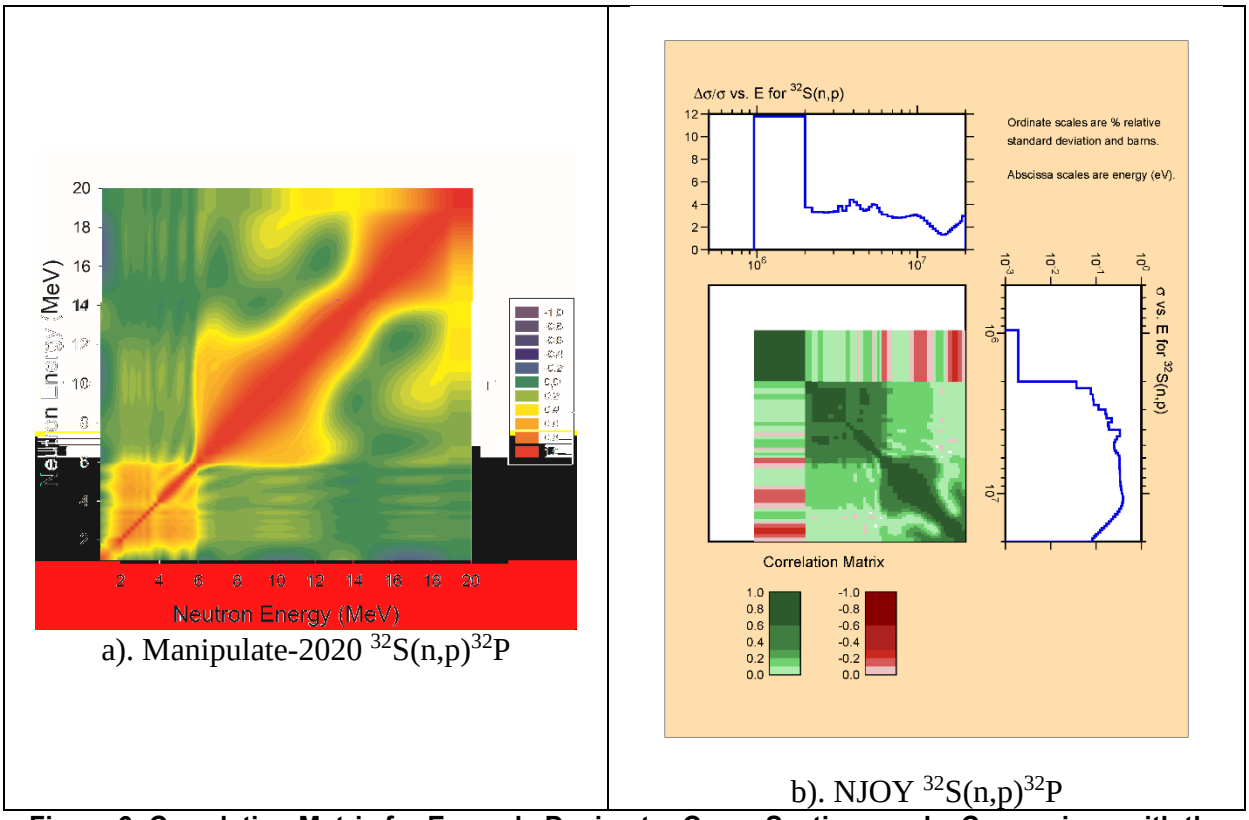
b). NJOY $^{59}Co(n, \gamma)^{60}Co$



a). Manipulate-2020 $^{109}Ag(n,\gamma)^{110m}Ag$



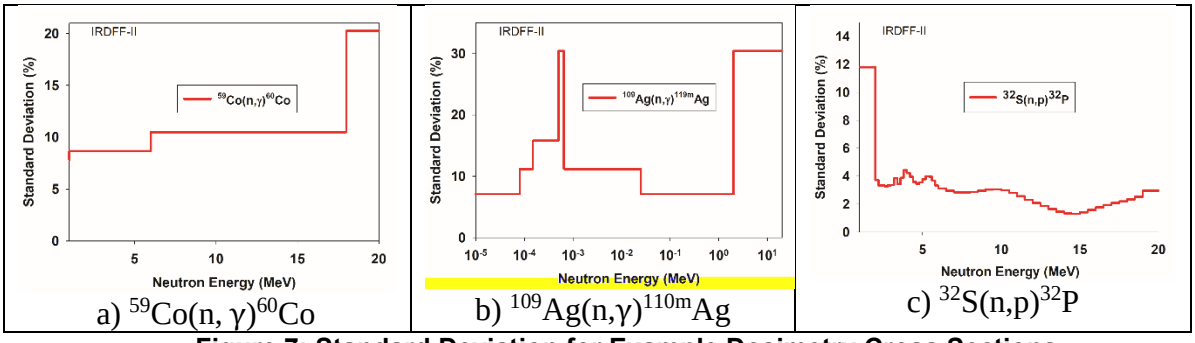
b). NJOY $^{109}Ag(n,\gamma)^{110m}Ag$



a). Manipulate-2020 $^{32}S(n,p)^{32}P$

b). NJOY $^{32}S(n,p)^{32}P$

Figure 6: Correlation Matrix for Example Dosimetry Cross Sections and a Comparison with the NJOY Output



a) $^{59}Co(n,\gamma)^{60}Co$ b) $^{109}Ag(n,\gamma)^{110m}Ag$ c) $^{32}S(n,p)^{32}P$

Figure 7: Standard Deviation for Example Dosimetry Cross Sections

6.8. Enforce Positive Definite Attribute and Unit Normalization for the Neutron Spectrum Covariance Matrix

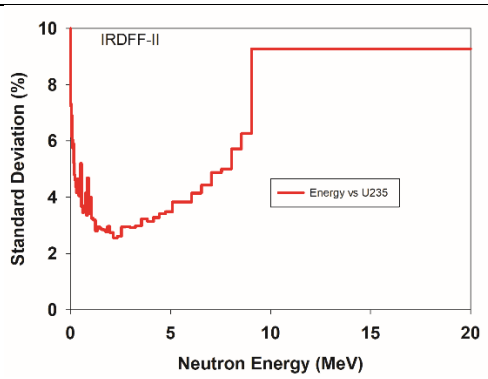
6.8.1. Description

6.8.2. Output

6.9. Spectrum Correlation Matrix Conversion for Plotting Interface

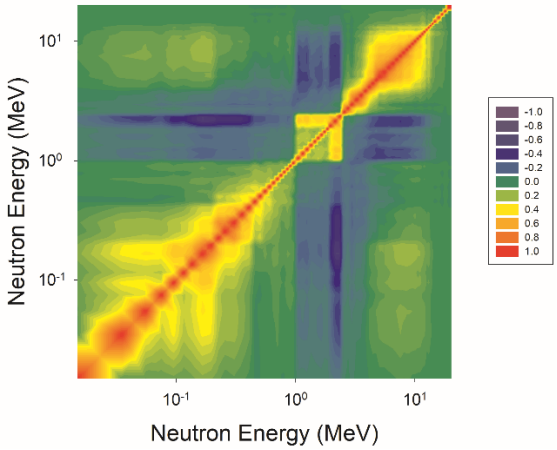
6.9.1. Description

6.9.2. Output

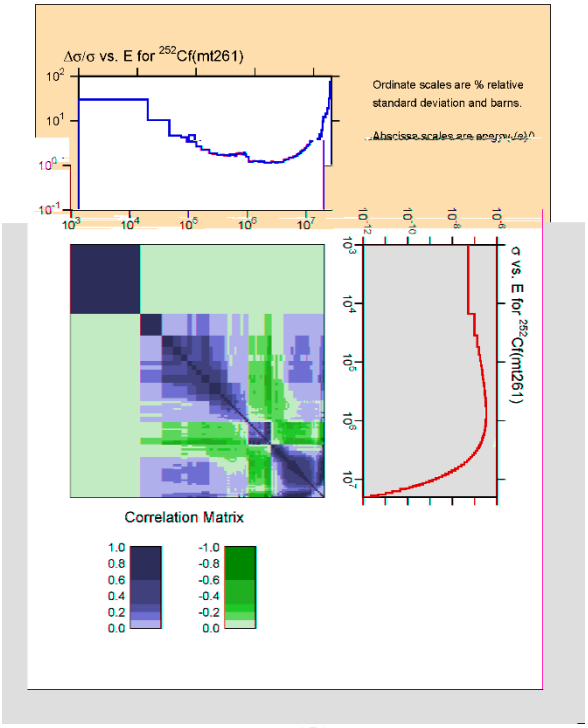


a) ^{252}Cf(s.f.) b) ^{235}U(n_{th})

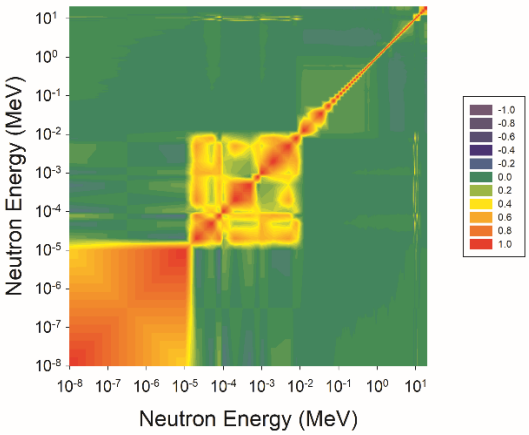
Figure 8: Standard Deviation for Benchmark Field Neutron Spectra



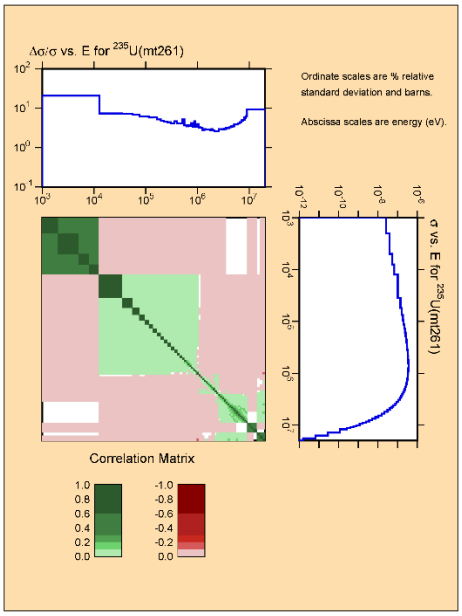
a). Manipulate-2020 ^{252}Cf(s.f.)



b). NJOY ^{252}Cf(s.f.)



a). Manipulate-2020 ^{235}U(n_{th})



b). NJOY ^{235}U(n_{th})

Figure 9: Correlation Matrix for Example Neutron Benchmark Fields and a Comparison with the NJOY Output

6.10. Spectrum Uncertainty in Spectrum-Averaged Response

6.10.1. Description

6.10.2. Output

Table 6-3. Summary of Spectrum Uncertainty Characteristic of Spectrum/Cross Section Folding

Cross Section	Neutron Spectrum	<σ> (mb) [640-grp]	<σ> (mb) [cov-grp]	Spct. Unc. %
$^{59}Co(n,\gamma)^{60}Co$	$^{252}Cf(s.f.)$	4.864277	5.171	2.471
	$^{235}U(n_{th})$	5.026129	4.9889	0.7529
$^{109}Ag(n,\gamma)^{110m}Ag$	$^{252}Cf(s.f.)$	9.34751	9.280	0.5645
	$^{235}U(n_{th})$	9.739893	9.6348	0.7875
$^{32}S(n,p)^{32}P$	$^{252}Cf(s.f.)$	73.99146	75.171	0.7412
	$^{235}U(n_{th})$	66.82598	67.909	1.156

6.11. Cross Section Uncertainty in Spectrum-Averaged Response

6.11.1. Description

6.11.2. Output

Table 6-4. Summary of Response Uncertainty Characteristic of Spectrum/Cross Section Folding

Response / Cross Section	Neutron Spectrum	<σ> [640-grp]	Resp. Unc. %
$^{59}Co(n,\gamma)^{60}Co$	^{252}Cf(s.f.)	4.864277	4.078
	$^{235}U(n_{th})$	5.026129	3.996
$^{109}Ag(n,\gamma)^{110m}Ag$	^{252}Cf(s.f.)	9.34751	7.554
	$^{235}U(n_{th})$	9.739893	7.476
$^{32}S(n,p)^{32}P$	^{252}Cf(s.f.)	73.99146	2.484
	$^{235}U(n_{th})$	66.82598	2.534

7. CONCLUSION

REFERENCES

[ASTM1855] ASTM E1855-20, Standard Test Method for Use of 2N2222A Silicon Bipolar Transistors as Neutron Spectrum Sensors and Displacement Damage Monitors, Annual Book of ASTM Standards, Vol. 12.02, ASTM International, West Conshohocken, PA, 2020.

[ASTM261] ASTM E261-20, Standard Practice for Determining Neutron Fluence, Fluence Rate, and Spectra by Radioactivation Techniques, Annual Book of ASTM Standards, Vol. 12.02, ASTM International, West Conshohocken, PA, 2020.

[ASTM265] ASTM E265-20, Standard Test Method for Measuring Reaction Rates and Fast-Neutron Fluences by Radioactivation of Sulfur-32, Annual Book of ASTM Standards, Vol. 12.02, ASTM International, West Conshohocken, PA, 2020.

[IRDFF] Trkov, A., Griffin, P.J., Simakov, S.P., Greenwood, L.R., Zolotarev, K.I., Capote, R., et al., "IRDFF-II: A New Neutron Metrology Library", Nuclear Data Sheets, Vol. 163, pp. 1-108, 2020.

[ENDF] "ENDF-6 Formats Manual: Data Formats and Procedures for the Evaluated Nuclear Data Files ENDF/B-VI, ENDF/B-VII and ENDF/B-VIII", edited by A. Trkov, M. Herman, D.A. Brown, report CSEWG Document ENDF-102, report BNL-203218-2018-INRE, February 1, 2018. Available at: https://www.nndc.bnl.gov/csewg/docs/endf-manual.pdf

[ENDF8a] D.A. Brown, M.B. Chadwick, R. Capote, et al., "ENDF/B-VIII.0: The 8th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data," Nuclear Data Sheets, Vol. 148, pp. 1-142, February 2018.

[ENDFB8b] ENDF/VI Release 8 files are available at:
http;://www.nndc.bnl.gov/csewg/summary8.html

Sandia National Laboratories

Sandia National Laboratories is a multimission laboratory managed and operated by National Technology & Engineering Solutions of Sandia LLC, a wholly owned subsidiary of Honeywell International Inc. for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525.